Gravitational Suppression of Quantum Decoherence
via Variable Vacuum Foam Density

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Abstract

We present a derivation of quantum decoherence rates within a vacuum pressure gradient model of gravity, in which local vacuum density decreases near massive objects according to rho_vac(r) = rho_0(1 - 2GM/rc²). This model predicts that decoherence rates are suppressed near massive objects — in direct opposition to the Diosi-Penrose (DP) model, which predicts enhancement. The sign reversal in the gravitational correction term produces a measurable difference of 3GM/rc² in decoherence rate ratios at two gravitational potentials. We additionally identify that quantum decoherence and macroscopic void decay are the same physical process at different scales — both governed by local vacuum bubble density eroding void-pair boundaries. We propose a concrete experimental protocol using atomic interferometry at varying gravitational potentials, achievable with existing space-based platforms, that can discriminate between this model and Diosi-Penrose. We further predict that qubit coherence times should be measurably longer at higher altitude, with a fractional improvement of 8.22×10^{-11} between Earth surface and the International Space Station — within reach of next-generation trapped ion experiments.*

Keywords: *quantum decoherence, vacuum energy density, gravitational effects on quantum coherence, Diosi-Penrose model, atom interferometry, void-pair conservation*

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1. Introduction

The relationship between gravity and quantum coherence remains one of the central unsolved problems in fundamental physics. Two competing models exist for how gravitational fields should affect quantum decoherence rates. The Diosi-Penrose (DP) model [1,2] predicts that decoherence rate increases with local gravitational field strength — matter in stronger gravitational fields should decohere faster. Recent experimental proposals [3] have begun to approach the sensitivity required to test this prediction.

We examine the opposite prediction arising from a vacuum foam model of gravity. If the local vacuum density decreases near massive objects — as required by consistency with the Schwarzschild metric — then the rate of vacuum-induced decoherence also decreases near mass. The two models differ by a sign in the gravitational correction term. That sign difference is experimentally distinguishable and constitutes a clean test of the underlying physics.

We additionally present a physical mechanism for why decoherence rate should scale with local vacuum density: quantum decoherence is void boundary erosion at the atomic scale, driven by the density of vacuum foam bubbles pressing on void-pair imprint boundaries. This mechanism unifies quantum decoherence with macroscopic void decay — both are the same process at different scales, both suppressed near mass, both governed by the local vacuum density equation.

2. Theoretical Framework

2.1 Vacuum Foam Density in a Gravitational Field

We adopt a model in which the vacuum is a structured medium — a quantized foam of Planck-scale bubble units — at baseline density:

```
rho_0 = m_Planck / l_Planck^3 = 5.155 x 10^96 kg/m^3      (1)
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In the presence of a gravitational source of mass M, the local vacuum density is displaced. Consistency with the Schwarzschild metric requires vacuum density to vanish at the Schwarzschild radius r_s = 2GM/c². The corrected vacuum density:

```
rho_vac(r) = rho_0 * (1 - 2GM/(r*c^2))      (2)
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Note that the factor of 2 is required: at r = r_s, equation (2) gives rho_vac = 0, consistent with the physical interpretation that the vacuum foam is completely evacuated at the event horizon. A factor of 1 instead of 2 gives rho_vac = rho_0/2 at r_s, which is incorrect. This correction propagates through all subsequent equations.

The vacuum equation of state, required for dimensional consistency of the gravity derivation:

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P_0 = rho_0 * c^2 = 4.633 x 10^113 Pa      (3)
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This is the equation of state of a maximally stiff medium in which pressure waves propagate at c.

2.2 Decoherence Rate from Vacuum Density

In standard quantum mechanics, decoherence arises from entanglement of a quantum system with environmental degrees of freedom. In the vacuum foam model, decoherence arises from a specific physical mechanism: the void-pair imprint — the record of a quantum measurement outcome written into the foam — is eroded by surrounding vacuum bubbles pressing on its boundary. The erosion rate is proportional to the local bubble density.

This gives a position-dependent decoherence rate:

$$\texttt{Gamma(r) = Gamma\_0 * rho\_vac(r) / rho\_0 = Gamma\_0 * (1 - 2GM/(r*c\^{}2))} \qquad \texttt{(4)}$$

and a corresponding position-dependent coherence time:

$$\texttt{tau(r) = tau\_0 / (1 - 2GM/(r*c\^{}2))} \qquad \texttt{(5)}$$

Near a massive object (r → r_s): Gamma → 0, tau → infinity. Decoherence is suppressed. Far from all mass (r → infinity): Gamma → Gamma_0, tau → tau_0. This is the undisturbed foam decoherence rate.

2.3 Sign Reversal Relative to Diosi-Penrose

The Diosi-Penrose model predicts:

$$\texttt{Gamma\_DP(r) / Gamma\_DP(inf) = 1 + GM/(r*c\^{}2)} \quad \texttt{[enhanced near mass]} \qquad \texttt{(6)}$$

The vacuum foam model predicts:

$$\texttt{Gamma\_VF(r) / Gamma\_VF(inf) = 1 - 2GM/(r*c\^{}2)} \quad \texttt{[suppressed near mass]} \qquad \texttt{(7)}$$

The difference between the two predictions at gravitational potential phi = GM/rc^2:

$$\texttt{Delta(Gamma\_DP - Gamma\_VF) / Gamma\_0 = 3 * GM / (r * c\^{}2) = 3 * phi} \qquad \texttt{(8)}$$

This difference is 3phi — three times the gravitational potential. At Earth's surface phi = 6.96×10^{-10}. The fractional difference between the two model predictions is 2.09×10^{-9}. This is small but non-zero and grows with gravitational potential, making it measurable with sufficiently sensitive experiments.

2.4 Decoherence as Void Decay — A Unification

We note that the decoherence rate equation (4) is identical in form to the macroscopic void decay equation:

$$\texttt{dV\_void/dt = -k * rho\_vac(r) * A\_boundary} \qquad \texttt{(9)}$$

where k is the void erosion constant [$m^4/(kg \cdot s)$], and A_boundary is the void surface area. Both quantum decoherence and macroscopic void decay are governed by local vacuum density eroding a void boundary — one at atomic scale, one at macroscopic scale. This unification implies that sonoluminescence (macroscopic void collapse), quantum decoherence (atomic void-pair imprint erosion), and gravitational effects on coherence time are all manifestations of the same underlying process.

3. Numerical Predictions

3.1 Earth Surface vs International Space Station

We compute the predicted fractional improvement in coherence time between Earth surface and the International Space Station at altitude h = 400 km.

Gravitational potential difference:

```
delta_phi = GM/R_E - GM/(R_E + h) = 4.11 x 10^-11      (10)
```

Fractional coherence time improvement (vacuum foam model):

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delta_tau / tau_0 = 2 * delta_phi = 8.22 x 10^-11      (11)
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Absolute improvement for different qubit technologies:

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Superconducting qubits (tau_0 ~ 1 ms):  delta_tau ~ 0.08 ps  [below current sensitivity]

Trapped ion qubits  (tau_0 ~ 100 s):   delta_tau ~ 8.2 ns   [within reach of precision experiments]
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The Diosi-Penrose model predicts the opposite sign: coherence time should be shorter at higher altitude by a comparable magnitude. The two models therefore make predictions that differ by approximately $2 \times 8.22 \times 10^{-11} = 1.64 \times 10^{-10}$ fractionally — the combined separation of the two predictions.

3.2 Scaling with Gravitational Potential

The effect scales linearly with gravitational potential difference. Experiments at higher gravitational potential difference — lunar surface vs orbit, or deep gravitational wells — would produce proportionally larger signals. For a laboratory at altitude h_2 vs sea level $h_1 = 0$:

```
delta_tau / tau_0 = 2 * G * M_E / c^2 * (1/R_E - 1/(R_E + h))      (12)
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At h = 1 km altitude: delta_tau/tau_0 = 2.18×10^{-13}. This is below current sensitivity but establishes the gradient prediction for future experiments.

4. Experimental Protocol

4.1 Discrimination Test

The minimal experimental requirement is a measurement of coherence time tau at two different gravitational potentials phi_1 < phi_2 (phi_2 at greater height). Measure the ratio:

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R = tau(phi_2) / tau(phi_1)      (13)
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Vacuum foam model prediction: R > 1 (coherence time longer at higher altitude).

Diosi-Penrose model prediction: $R < 1$ (coherence time shorter at higher altitude).

A measurement of R with sufficient precision to determine its sign unambiguously constitutes a discrimination between the two models. The required precision is approximately 10^{-10} in fractional coherence time — within the projected capability of BECCAL [4] and STE-QUEST [5].

4.2 Recommended Platform

Trapped ion qubits are the preferred platform. Their coherence times of order 100 seconds produce absolute timing differences of approximately 8 nanoseconds between Earth surface and ISS altitude. This is more accessible than superconducting qubits at 1 millisecond coherence, which produce differences of order 0.08 picoseconds.

The experiment requires: (1) identical trapped ion qubit systems at two altitudes; (2) coherence time measurement precision better than 1 nanosecond per 100 seconds; (3) systematic error budget that accounts for magnetic field differences, temperature differences, and vibrational noise between the two sites. The gravitational effect is the residual after all other systematic differences are subtracted.

Space-based platforms avoid ground vibration noise and allow precise altitude control. The BECCAL facility on the International Space Station provides a suitable high-altitude platform. A complementary ground station provides the reference measurement.

4.3 Additional Test — Void Decay Signature

The unification of decoherence with void decay (Section 2.4) predicts a characteristic emission signature at each decoherence event — the foam's response to rapid void boundary erosion. This signature should be consistent in spectral profile across quantum decoherence events, sonoluminescence collapse events, and cavitation erosion events. If measurable, it constitutes an independent confirmation of the shared underlying mechanism, distinct from the coherence time measurement.

5. Discussion

The vacuum foam model and the Diosi-Penrose model represent two physically distinct pictures of the relationship between gravity and quantum coherence. In DP, gravity actively promotes collapse of quantum superposition — stronger gravity means faster decoherence. In the vacuum foam model, gravity reduces the local vacuum density that drives decoherence — stronger gravity means slower decoherence. These are not minor quantitative differences but qualitatively opposite predictions arising from fundamentally different conceptions of what causes quantum-to-classical transition.

The sign difference is experimentally distinguishable. Both models predict effects of comparable magnitude at accessible gravitational potential differences. The experiment is therefore not merely a precision measurement of a known effect but a qualitative discrimination between two competing physical mechanisms.

The identification of decoherence as void boundary erosion (Section 2.4) provides a physical mechanism that standard decoherence theory does not offer. Standard environmental decoherence theory (Zurek [6], Joos and Zeh [7]) correctly predicts that macroscopic objects decohere rapidly and that environmental coupling drives the quantum-to-classical transition. The vacuum foam model is consistent with this — the environment creates void-pair imprints — but adds the substrate mechanism: the imprints are eroded by local vacuum bubble density, which is why the rate depends on gravitational potential.

We note that the covariant derivation of equation (2) from quantum field theory in curved spacetime remains open work. The equation is consistent with the Schwarzschild metric and with established gravitational physics, but a full derivation from first principles within a covariant framework would strengthen the theoretical foundation considerably. We regard this as the primary open theoretical question for this model.

6. Conclusion

We have presented a vacuum foam density model of quantum decoherence that predicts gravitational suppression of decoherence rates — the opposite sign to Diosi-Penrose. The key predictions are: (1) coherence time increases with altitude, fractionally by 8.22×10^{-11} between Earth surface and ISS; (2) the sign of this effect discriminates between the two models; (3) decoherence and macroscopic void decay are the same physical process governed by the same equation. These predictions are testable with next-generation space-based atom interferometry platforms. The experiment requires no new technology — only sufficient precision in coherence time measurement at two altitudes to determine the sign of the gravitational correction.

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References

[1] Diosi, L. (1987). A universal master equation for the gravitational violation of quantum mechanics. Physics Letters A, 120(8), 377–381.

[2] Penrose, R. (1996). On gravity's role in quantum state reduction. General Relativity and Gravitation, 28(5), 581–600.

[3] Oppenheim, J., Sparaciari, C., Šoda, B., & Weller-Davies, Z. (2023). Gravitationally induced decoherence vs space-time diffusion. Nature Communications, 14, 7910.

[4] Frye, K., et al. (2021). The Bose-Einstein Condensate and Cold Atom Laboratory. EPJ Quantum Technology, 8, 1.

[5] Aguilera, D. N., et al. (2014). STE-QUEST — Test of the Universality of Free Fall Using Cold Atom Interferometry. Classical and Quantum Gravity, 31, 115010.

[6] Zurek, W. H. (2003). Decoherence, einselection, and the quantum origins of the classical. Reviews of Modern Physics, 75, 715.

[7] Joos, E., & Zeh, H. D. (1985). The emergence of classical properties through interaction with the environment. Zeitschrift für Physik B, 59(2), 223–243.

[8] Schwarzschild, K. (1916). Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie. Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften, 189–196.

[9] Planck, M. (1899). Über irreversible Strahlungsvorgänge. Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften, 5, 440–480.